

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Technical Presentation

Course Code:	CSA0309	Course Title:	Data Structures
CO Assessed:	CO1 – Imitation (BL3, BL4)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	35% of CIA	Total Marks:	100
CO1 Statement:	Apply suitable data structures and ADTs for efficient problem solving by analyzing time and space complexity.		
Student Name:		Reg. No.:	
Section / Batch:		Date:	13/07/2026

Code of Conduct

I _____ PRENA M[192524056] _____
(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.
Signature of the Student: _____ PRENA M _____

Instructions

- This assessment contains technical presentation. Total: 100 Marks.
- When a student delivers a technical presentation on a data structures topic, evaluation should be based on the following requirements: Concept explanation, Real-world application and Clarity of presentation. •
Demonstrate clear understanding, not just reading slides
- Use of tools: PowerPoint / Google Slides.
- Duration: 7–10 minutes presentation + 3 minutes Q&A.

Section / Topic	Focus	Q Nos	Marks
Data Structure	Real-Time Applications	1	100
GRAND TOTAL:			100 Marks

Annexure A – Technical Presentation

A web browser supports navigating backward through previously visited pages. Recommend a suitable ADT and prepare a technical presentation explaining how it improves performance efficiency. (100 Marks)

(OR)

A music streaming application maintains a playlist where songs can be added, removed, or reordered dynamically. Recommend a suitable data structure and justify based on flexibility and efficiency. Prepare a technical Presentation. (100 Marks) (OR)

A metro rail network maps stations and routes to determine reachable stations from a source station. Suggest a suitable data structure for representation and justify your answer. (100 Marks)

(OR)

A calculator application evaluates arithmetic expressions entered by users. Recommend a suitable data structure and justify based on expression processing efficiency. (100 Marks) (OR)

A cafeteria token system serves customers strictly in the order of arrival. Identify the suitable ADT and justify why it ensures fairness. (100 Marks)

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	20	Demonstrates deep and accurate understanding of data structure with insights and connections	Shows clear understanding with minor gaps	Basic understanding; some misconceptions present	Limited or incorrect understanding
Content Organization	30	Logical, wellstructured, smooth flow; strong introduction and conclusion	Mostly organized with clear flow	Some organization but lacks clarity or transitions	Disorganized, difficult to follow
Technical Depth	20	Includes algorithms, complexity analysis, and advanced insights	Includes algorithm and basic complexity analysis	Limited technical details; missing depth	Minimal or no technical content
PowerPoint Slides	20	Excellent design, clear visuals, enhances understanding	Good slides with minor issues	Basic slides; somewhat cluttered or unclear	Poor slides or no visual support
Communication Skills	10	Clear, confident, engaging delivery; excellent articulation	Clear delivery with minor hesitation	Some hesitation; unclear in parts	Poor communication; difficult to understand
Faculty In-charge		Course Coordinator		Program Director	

Signature: _____

Date: _____

Signature: _____

Date: _____

Signature: _____

Date: _____

TECHNICAL PRESENTATION

Efficient Arithmetic Expression Evaluation in Calculator Applications

A Stack-Based Approach to Parsing and Evaluating Expressions

GUIDED BY

DR. UMA PRIYADHARSHINI

PRESENTED BY

PRENA M
192524056

Abstract

Calculator applications must evaluate arithmetic expressions that involve operator precedence, nested parentheses, and left-to-right associativity — all while returning a result instantly. This presentation examines that problem and recommends the stack (LIFO) as the data structure best suited to the task. We define the requirements an evaluation engine must satisfy, compare the stack against arrays, queues, and trees, and walk through the two core algorithms it enables: converting infix expressions to postfix using the shunting-yard method, and evaluating the resulting postfix expression. A full worked example and a time/space complexity analysis show that the stack achieves $O(n)$ time with $O(1)$ per-token operations — outperforming naive re-scanning and matching the efficiency of tree-based parsing with far less overhead.

Stack (LIFO)

Shunting-Yard

Postfix Evaluation

$O(n)$ Time Complexity

Expression Trees

INTRODUCTION

Calculators, Made to Think in Order

From phone apps to spreadsheet formula bars, calculators are among the most-used pieces of software in existence. Every one of them faces the same underlying challenge: turning a flat string of characters typed by a user into a single correct numeric answer, respecting the rules of arithmetic along the way.

The choice of data structure behind that process directly determines how fast, reliable, and scalable the calculator is. This presentation walks through that choice — from the problem, to the recommended structure, to the algorithms and performance behind it.



The Problem

What expression evaluation demands



The Recommendation

Why the stack (LIFO) fits best



The Algorithms

Infix $\rightarrow$ postfix and postfix evaluation



The Evidence

Efficiency analysis and real-world use



Mevalurkuppam, Tamil Nadu,
India 🇮🇳

22g8+9hp, Mevalurkuppam, Tamil Nadu 602105,
India

Lat 13.026278° Long 80.016283°

Friday, 24/07/2026 02:57 PM GMT +05:30

GPS Map Camera